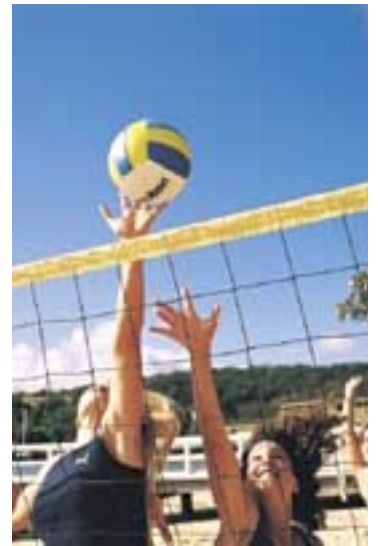


June, 2001

MMS

Multimedia Messaging Service



Share the good times

Contents

Contents	2
Preface	3
Purpose of this document	3
Introduction	3
Meet Multimedia Messaging	3
The messaging market	4
Evolution	4
SMS	4
EMS	4
MMS	4
Demand	4
MMS content	5
Text	5
Graphics	5
Audio	5
Images	6
Video	6
SMIL presentations	6
User scenario 1 - "Businessman on the move with MMS"	6
User scenario 2 - "Susan, 22-year-old traveller"	6
Benefits	7
End users	7
Network operators	7
Service and content providers	8
Technical features	8
Architecture	8
Message conversion	9
OTA configuration	9
Messaging technologies – how they complement each other	9
SMS - text messaging	9
EMS - enhanced messaging	9
MMS - full digital context exchange	10
E-mail	10
Related information	10
Links	10

This White Paper is published by:

Ericsson Mobile Communications AB
SE-221 83 Lund, Sweden

Phone: +46 46 19 30 00
Fax: +46 46 19 41 82
www.ericsson.com

First edition (June 2001)
Publication number: EN/LZT 1084963R1A

This document is published by **Ericsson Mobile Communications AB**, without any warranty.

Improvements and changes to this text necessitated by typographical errors, inaccuracies of current information or improvements to programs and/or equipment, may be made by **Ericsson Mobile Communications AB** at any time and without notice. Such changes will, however, be incorporated into new editions of this document. Any hard copies of this documents are to be regarded as temporary reference copies only.

Preface

Purpose of this document

The Ericsson MMS White Paper is designed to give the reader an understanding of how MMS works as the ultimate messaging application.

People who can benefit from this document include:

- Corporate buyers
- IT professionals
- Software developers
- Support engineers
- Business decision-makers

More information, useful for product, service and application developers, is published on the Ericsson Developers' Zone. The site at <http://www.ericsson.com/developerszone/> contains up-to-date information about technologies, products and tools.



Introduction

Meet Multimedia Messaging

To meet the 3GPP standard Multimedia Messaging Service (MMS) is to know complete communication. With virtually no limit to the content that can be transmitted mobile-to-mobile or network-to-mobile, MMS offers total freedom to convey ideas, to supply or exchange information or to express oneself. Using MMS is an easy way to express feelings.

Multimedia Messaging technology represents the pinnacle of the Ericsson mobile messaging evolution. Beginning with the enormously popular Short Message Service (SMS) for simple text messages, the exciting Enhanced Message Service (EMS) for illustrated text messages with sound has entered the market as the second step. MMS will become the ultimate messaging application, allowing users to create unique messages, using all types of multimedia.

The messaging evolution fosters steady expansion of the marketplace for network operators and service providers. Studies show that users are willing to pay more for multimedia messages than for ordinary text messages. There is virtually no limit to the kind of services that network operators and service providers can offer users. Subscriptions to the comic strip of the day, weather forecasts in pictures... A picture says more than a thousand words and is more fun to look at.

Ushering in the new generation of mobile telephony by paving the way for an array of groundbreaking applications, Multimedia Messaging technology will result in users expecting more from their applications and developers continuing to meet these demands. MMS and the "face to face" contact it provides will simply become synonymous with mobile communication.

Using the Wireless Application Protocol (WAP) as bearer technology and powered by the high-speed transmission technologies EDGE, GPRS and UMTS (W-CDMA), Multimedia Messaging allows users to send and receive messages that look like PowerPoint-style presentations. The messages may include any combination of text, graphics, photographic images, speech and music clips or video sequences. MMS will serve as the default mode of messaging on all terminals, making total content exchange second nature. From utility to sheer fun, it offers benefits at every level and to every kind of user.

The messaging market

Building on proven success, Ericsson paves the way to a complete communication experience.

The messaging market is characterized by the intentional evolution of progressively advanced services. The goal of evolution is to ease the transition of total multimedia communication for users, operators and content providers. Driving the market is a steadily growing, enthusiastic consumer demand that gives content and service providers the confidence to embrace third generation systems.

Evolution

As the foremost provider of end-to-end telecommunications solutions, Ericsson remains committed to fostering a timely and seamless evolution of the messaging market. This approach is both user- and operator-friendly, as it leads the market towards full mobile multimedia. The key stages of the evolution entail the sequential release of the following services:

- Short Messaging Service (SMS) – text messaging
- Enhanced Messaging Service (EMS) – illustrated text messaging
- Multimedia Messaging Service (MMS) – full multimedia content exchange

SMS

The Short Messaging Service (SMS) was launched in 1992 and has become the most successful wireless data service to date. SMS allows mobile phone users to send and receive text messages of up to 160 characters in a cost- and time-efficient manner. SMS is a “store and forward” service, meaning that messages are not sent directly between users but via an SMS centre. This aspect allows for a number of key SMS attributes, such as instant delivery, nominal tariffing and message delivery unhindered by network traffic. The “store and forward” service also allows simultaneous SMS and voice capability and international “roaming” without international fees. SMS is also used to notify users of incoming e-mail, voice mail or faxes, as well as to inform them about weather forecasts, news headlines, stock quotes or other events they can subscribe to.

EMS

EMS can be referred to as “enhanced SMS”, and adds life to the users’ SMS text messages. Messages sent with the

3GPP standard EMS (Enhanced Message Service) contain a combination of text and simple pixel-image and/or melody. Users may download images and melodies from the Internet, or for even greater self-expression, create them on their own directly in the phone. Unlike SMS messages, the text of an EMS message can be formatted using a variety of fonts, sizes, type styles, etc. Ericsson's EMS is a backwards-compatible service, meaning that the text portion of its messages can be received by terminals not supporting EMS. EMS is an open 3GPP standard, and paves the way for the introduction of MMS.

MMS

MMS is the pinnacle of the messaging evolution. Currently being defined and specified by 3GPP as a standard for third generation implementation, the Multimedia Messaging Service (MMS) completes the potential of messaging. MMS is expected to become the preferred messaging method of mobile terminal users, since there are virtually no limits to the content of an MMS transmission. An MMS message can contain formatted text, graphics, data, animations, images, audio clips, voice transmissions and video sequences. Sending digital postcards and PowerPoint-style presentations is expected to be among the most popular user applications of MMS. Greatly anticipated by young users in particular, MMS is projected to fuel the growth of related market segments by as much as forty percent.



Demand

SMS has been an unqualified success for both end users and service providers. With practically no high-profile marketing undertaken by mobile terminal vendors or network operators, this value-added service is currently used to send some 15 billion messages per month worldwide (January 2001). The number of messages sent is continuously increasing.

An extremely positive indicator of the emerging market is the fact that young users in particular have driven messaging growth. In parts of Scandinavia, for example, more than half of a young person's mobile phone bill

consists of SMS charges. These users send over 100 messages per month, and as the capabilities of mobile terminals continue to get better and better (bigger displays, built-in cameras, higher sound quality), the frequency of sending messages will increase substantially.

A strong demand for Multimedia Messaging exists. Not surprisingly, it has been found that the younger the user, the greater his or her interest is in MMS services. As clear evidence of the huge market potential for operators and service providers, studies have found that users expect and accept that the richer the content, the more they will be charged for it. While users want to retain the ability to choose among the different messaging services (SMS, EMS and MMS), they consider the rich content of multimedia messages to be five times more valuable than simple text messages.



Figure 1. An MMS message can contain images, music, video and audio, and graphics.

MMS content

Rich content to communicate the richness of our lives.

Although MMS is a direct descendant of SMS, the difference in content is dramatic. The size of an average SMS message is about 140 bytes, while the average size of an MMS message will (in the early stages) be around 30,000 bytes, but is actually unlimited. In the future, the user will be able to store a large number of messages, including those with video clips. The size of these messages will be about 100,000 bytes. That is why the key word to describe MMS content is rich. Complete with words, sounds and images, MMS content is endowed with the user's ideas, feelings and personality. An MMS message can contain one or more of the following:

Text

As with SMS and EMS, an MMS message can consist of normal text. The length of the text is unlimited, and in the future it will be possible to format the text. The main difference between an EMS and MMS message is that in an MMS message, text can be accompanied not only by simple pixel images or melodies but by photographic images, graphics, audio clips and video sequences.

Graphics

Graphs, tables, charts, diagrams and layouts are just a few examples of the kinds of MMS graphic capabilities sure to have a major impact on the way we work. Maps, drawings, sketches and animations are likely to play a larger part in our personal lives, helping us to find our way, feel safe, express ourselves and have fun. MMS supports animated GIFs.

Audio

MMS provides the ability to add full sound to a message. Not only can users share a favourite song with a friend, but they can use the mobile phone to record sound and send it along with a message. Because sound includes speech as well as music, this extra dimension of an MMS message makes for tremendously enhanced immediacy of expression and communication. Rather than sending a downloaded birthday jingle in EMS, for example, a user can send a clip of his or her own personal rendition of "Happy Birthday". With MMS in a mobile phone, the user can download MP3 files, and the MMS standard also supports streaming of sound as well as images.

Images

By using either a digital camera attached to the mobile terminal with a cable or a built-in digital camera, users can take a snapshot and immediately send it to a recipient. The ability to send images is one of the most exciting attributes of MMS, as it allows users to share meaningful moments with friends, family and colleagues. Mobile image transmission also offers inestimable utility in business applications, from sending on-site pictures of a construction project to capturing and storing an interesting design concept for later review. Editing an image by adding text allows users to create their own electronic postcards, an application that is expected to substantially cut into the traditional postcard-sending market.

Video

The ultimate extension of MMS's digital imaging capabilities, MMS video content, once developed, could initially comprise something like 30-second video clips. Instead of using, for example, the mobile device's digital camera and media editor to photograph a scene, label it with text and add appropriate audio, users will be able to record the scene and transmit the clip to a recipient. In the future, streaming of video clips will be possible. This will be a popular feature for people subscribing to news and entertainment services. The list of possible applications of this extremely exciting type of MMS content is virtually endless.

SMIL presentations

Standing for Synchronized Multimedia Integration Language and pronounced "smile", SMIL allows for the creation and transmission of PowerPoint-style presentations on the mobile device. SMIL is an advanced XML-based protocol, and Ericsson MMS supports a subset of this protocol. Using a simple media editor, users can incorporate audio and video along with still images, animations and text to assemble full multimedia presentations.

The idea of SMIL is to allow the user to customize the page timing in Powerpoint presentations. The user can decide in which order the image and text will be displayed, as well as for how long the images and text lines are to be shown in the display.

User scenario 1 - "Businessman on the move with MMS"

While on the train to work, Martin van Buren, a German-based Belgian account executive, choral music enthusiast and connoisseur of wieners, decides to play hooky from work and visit the Hanover Expo. He wants to use his Ericsson MMS-enabled mobile phone to call his secretary and inform her of his sudden

illness, but he doesn't want her to hear the background noise in the train compartment. So Martin sends his secretary an MMS message, appending a pixel image of a sad face to his text message, "too sick to work today".

Although he has read that the Expo is a dismal financial failure and the queues are horrendous, Martin has also heard that the wieners there are really delicious. To build up an appetite, he makes his way through several of the pavilions. In the Japanese exhibit, he sees the latest model of a stylish fountain pen, photographs it with the digital camera on his Ericsson phone, and immediately sends the image to his wife with the message, "Honey, this is what I want for Christmas". In the Russian exhibit, he hears the most divine choir music he has ever heard. Using the MMS audio capabilities on his mobile phone, he records a sample of the music and stores it for his later listening pleasure.

Alas, Martin's fun is interrupted by an urgent SMS message from his secretary. The boss wants to see the layouts Martin has been working on for the past week. He knew he should have turned the darned phone off! Oh well, it's no problem - he has the layouts with him in his briefcase. He finds a free table, spreads out the layouts and photographs them with his mobile phone. Then, to really dazzle the boss (and because he feels guilty now about skipping work), he uses the phone's media editor and some of the Russian choral music to create a full MMS message. He sends the SMIL presentation to his boss and feels much better.

Martin finally reaches the wiener tent. The wait has been worth it. They truly are the tastiest wieners he has ever had. If only he could use his Ericsson MMS-enabled phone to send one to a friend... He quickly browses the Internet to download a picture of a wiener. He finds a really good one and sends it off to one of his colleagues. Technology has a limit after all...

User scenario 2 - "Susan, 22-year-old traveller"

Sitting in her favourite restaurant in Tokyo, Susan, an American 22-year-old student of arts, sips her drink and stretches out her legs after an exhausting day of adventures. Listening to the wonderful, relaxing music that is filling the room, she suddenly spots this extremely handsome Japanese guy at the bar. She quickly slips her Ericsson MMS-enabled mobile phone out of her pocket and takes his picture. She taps a short message to her friend at the hotel, "Please come down, otherwise you're really missing out on something...". She encloses the picture of the guy and sends away her message, hoping that her friend will join her in a minute.

Just finished ordering another drink, she gets an MMS message from her local music store in L.A. The latest CD of her favourite singer is now being presented in the phone. She studies the cover, listens to the first two songs, and decides that she just has to have this one. She enters her credit number and sends away her order.

Her friend now comes dashing down the stairs and joins her at

the table. Unfortunately the Japanese guy is gone, but very soon they get something really interesting to set their eyes on. Susan's subscription to the latest sports events now appears on her Ericsson MMS-enabled mobile phone. Together the two friends study the video clips of the last 10 thrilling seconds of the football game with their favourite team. They get a summary of the commentary played back to them, and they can see some statistics on the display. Their favourite players won the game, and the two girls make their plans for tomorrow.



Benefits

Multimedia Messaging represents a classic win-win scenario - if it's good for the user, it's good for service and content providers.

Essentially enabling the mobile terminal to serve as image processor and conveyor, Multimedia Messaging accommodates the exchange of important visual information as readily as it facilitates fun. Business and leisure usage of MMS will be dynamically merged, resulting in enhanced personal efficiency for users and increased network activity for operators. In short, MMS affords total usage for total communication.

Because MMS uses WAP as its bearer technology and is being standardized by 3GPP, it has wide industry support and offers full interoperability, which is a major benefit to service providers and end users. Ease-of-use resulting from both the gradual steps of the messaging evolution and the continuity of user experience gained from interoperability is assured.

The MMS server, through which MMS messages are sent, supports flexible addressing (to both normal phone numbers (MSISDN) and e-mail accounts), which makes user interface more friendly and allows greater control for operators. The MMS server, moreover, is responsible for the instant delivery feature of MMS.

End users

The ability to take, edit and send a photographic image empowers users in all areas of life, enhancing personal connectivity through a more immediate sharing experience. It boosts efficiency in the workplace by serving as a powerful portable tool for both conveying and responding to ideas. Other applications, such as receiving a localized city map while on the road or a graph or layout while in a conference, enable users to "position" themselves in both their personal and business lives. With terminal-to-terminal as well as terminal-to-e-mail functionality, MMS can also serve as a virtual e-mail client, enabling users to circumvent many of the pitfalls of traditional e-mail service (cumbersome software, limited access outside the workplace, etc.).

Network operators

Multimedia Messaging promises increased air time, heavier all-around usage, service differentiation and customer loyalty - all key factors in increasing the revenue made from operating a network. Market studies have revealed that users are not only enthusiastic about MMS services but willing to pay as much as five times

more for them than they currently pay for SMS services. This cutting edge messaging platform takes full advantage of the high bandwidth technologies GPRS and WCDMA, which will result in high usage of MMS service. Investments in such technologies are not only warranted but wise, as they are destined to have short payback periods and enable operators to secure a strong market position early in the personal multimedia era.

Service and content providers

Multimedia Messaging applications will drive the continuous development of new services and content. The pure diversity of new services and demand for new content will ensure increased revenue, especially as service providers can offer content and charge for it much like they can for ringing tones today. Much like network operators, service and content providers stand to make great gains by developing for MMS functionality early, as they will secure a well-established position from which to cater to the demands of both operators and the public.

Technical features

The MMS standard, just like SMS, offers store-and-forward transmission (instant delivery) of messages, rather than a mailbox-type model. MMS is a person-to-person communications solution, meaning that the user gets the message directly into the mobile. He or she doesn't have to call the server to get the message downloaded to the mobile. Unlike SMS, the MMS standard uses WAP as its bearer protocol. MMS will take advantage of the high speed data transport technologies EDGE and GPRS and support a variety of image, video and audio formats to facilitate a complete communication experience.

Architecture

The MMS Centre (MMS-C) is comprised of the MMS Server, the MMS Proxy-Relay and the MMS Store. The MMS Centre is the central element of the MMS network architecture, providing storage and operational support, enabling instant delivery of multimedia messages from terminal-to-terminal and terminal-to-e-mail, and supporting flexible addressing. The centre's MMS Proxy-Relay interacts with the application being run on the MMS-enabled terminal to provide various messaging services. WAP is used as bearer of an MMS message between the MMSC and the MMS client (application). The WAP Gateway is used for delivery and retrieval of messages.

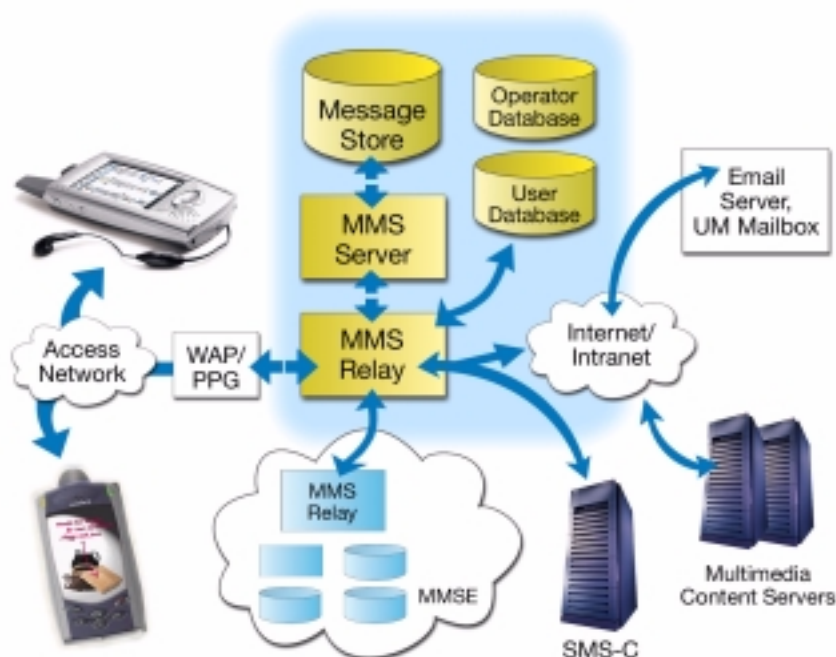


Figure 2. The architecture of MMS

Message conversion

The MMS-C is able to perform limited message conversion - for example, from MMS to SMS - so that processing and air time is not wasted in sending messages to mobile terminals that do not have adequate capability to receive them. It also handles service aspects such as store and forward, guaranteed delivery, subscriber preferences, operator constraints, and billing information. The MMS-C also vouches for high quality messaging, e.g. by format conversion. This means that the MMS-C recognizes which formats are supported in the mobile phone, and adapts the MMS messages to these formats.

OTA configuration

Users can easily get MMS into their phone. MMS supports OTA, meaning that the user does not have to configure the settings manually. The configuration is done by the operator. Supported formats

Currently being standardized, MMS is likely to support the following formats.

- Image - JPEG and GIF 87, 89a, WBMP
- Video coding - ITU-T H.263, MPEG 4 (simple profile),
- Audio - MP3, MIDI, AMR/EFR (for speech)
- Video - MPEG 4, H263



Messaging technologies – how they complement each other

Mobile communication is gradually becoming less textual and more visual as we move from simple text messages to multimedia messages. SMS will be upgraded to EMS (Enhanced SMS) in all Ericsson mobile phones starting with the T29s and T20e. MMS, the next generation of messaging, will be the standard messaging application in Ericsson mobile phones starting late 2001. E-mail is available already today in selected Ericsson products, and will continue to be an important messaging technology.

SMS - text messaging

Originally launched in 1992, SMS has become the most successful wireless data service. The messages are sent from mobile to mobile, via an SMS centre. From an operator's point of view, SMS is low-tech because minimal investment is needed to provide an efficient SMS service to subscribers and little maintenance is required. The size of an average SMS message is about 140 bytes.

EMS - enhanced messaging

At the beginning of 2001, EMS entered the market, adding new powerful functionality to the well-known SMS standard. With EMS, mobile phone users can add life to SMS text messaging in the form of pictures, animations, sound and formatted text. The beauty of EMS is that it uses existing SMS infrastructure and industry standards, keeping investment to a minimum for operators. It provides a familiar user interface and compatibility with existing phones and with other manufacturers. It is also a backward-compatible service, meaning that the text included in EMS messages can be received and viewed on mobile devices not supporting EMS. The EMS service will be a standard feature in Ericsson phones in the future and is a stable, complete and open standard, defined in 3GPP (the 3rd Generation Partnership Project).

MMS - full digital context exchange

At the end of 2001, Ericsson will launch their first terminal supporting MMS. As its name suggests, the Multimedia Messaging Service gives users the ability to send and receive messages consisting of a combination of text, sound, images and video. With person-to-person as well as person-to-Internet functionality, MMS can also serve as a virtual e-mail client.

Because MMS uses WAP as its bearer technology and is being defined and specified within 3GPP, it has wide industry support and offers full interoperability. MMS will require new network infrastructure (including an MMS Service Centre) and MMS-compliant terminals. In the MMS protocol, there is no limit of how much an MMS message can contain, but a typical MMS message will be around 30,000 bytes.

E-mail

E-mail, the dominating messaging technology in the fixed-line world, is already available to Ericsson mobile phone users. Selected Ericsson mobile phones are equipped with a fully fledged e-mail client, which gives access to the extensive e-mail world. By providing a WAP interface, Internet mail services go mobile, making it possible for people to read e-mails wherever they are.



Figure 3. SMS will be upgraded to EMS (Enhanced SMS) in all Ericsson mobile phones starting with the T29s and T20e. MMS, the next generation of messaging, will be the standard messaging application in Ericsson mobile phones starting late 2001. E-mail is available already today in selected Ericsson products, and will continue to be an important messaging technology.

Related information

Links

- <http://mobileinternet.ericsson.com/>
- <http://www.ericsson.com/wap/>
- <http://www.ericsson.com/mobilityworld/>
- <http://www.3gpp.org/>
- <http://www.wapforum.org/>

Index

A

Architecture
 MMS Centre 8
Architecture 8
Audio 5, 9

B

Benefits 7
 end users 7
 network operators 7
 service and content providers 8

D

Demand 4

E

E-mail 10
EMS 4
EMS - illustrated text messaging 9
End users 7
Evolution 4

G

Graphics 5

I

Image 9
Images 6

M

Message conversion 9
Messaging market 4
 demand 4
 EMS 4
 evolution 4
 MMS 4
 SMS 4
Messaging technologies 9
 e-mail 10
 EMS - illustrated text messaging 9
 MMS - full digital context exchange 10
 SMS text messaging 9
MMS 4
MMS - full digital context exchange 10
MMS Centre 8
MMS content 5
 audio 5
 graphics 5
 images 6
 SMIL presentations 6
 text 5
 video 6
MMSC 8

N

Network operators 7

O

OTA configuration 9

S

Service and content providers 8
SMIL presentations 6
SMS 4
SMS - text messaging 9
Supported formats 9

T

Technical features 8
 architecture 8
Text 5
the messaging market 4

U

User scenario 6

V

Video 6, 9
Video coding 9